

ASSIGNMENT ON WEB DESIGNING-III

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ROLL Number: BA23-447

Semester: BA 6th Sem

Subject: Vocational Training Course(VTC)

Assignment Number: 1

Title: Vlog Application - Create, Read, and Delete Operations
using React Router

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OBJECTIVE

This assignment aims to develop a full-featured vlog application using React and React Router that demonstrates CRUD operations (Create, Read, Delete). The primary objectives are to understand component-based architecture, implement client-side routing for seamless navigation, manage application state effectively, and persist data using browser local storage. Through this project, we aim to learn modern React best practices, CSS responsive design, and how to build scalable single-page applications (SPAs) that provide an excellent user experience across different devices.

TOOLS & TECHNOLOGIES

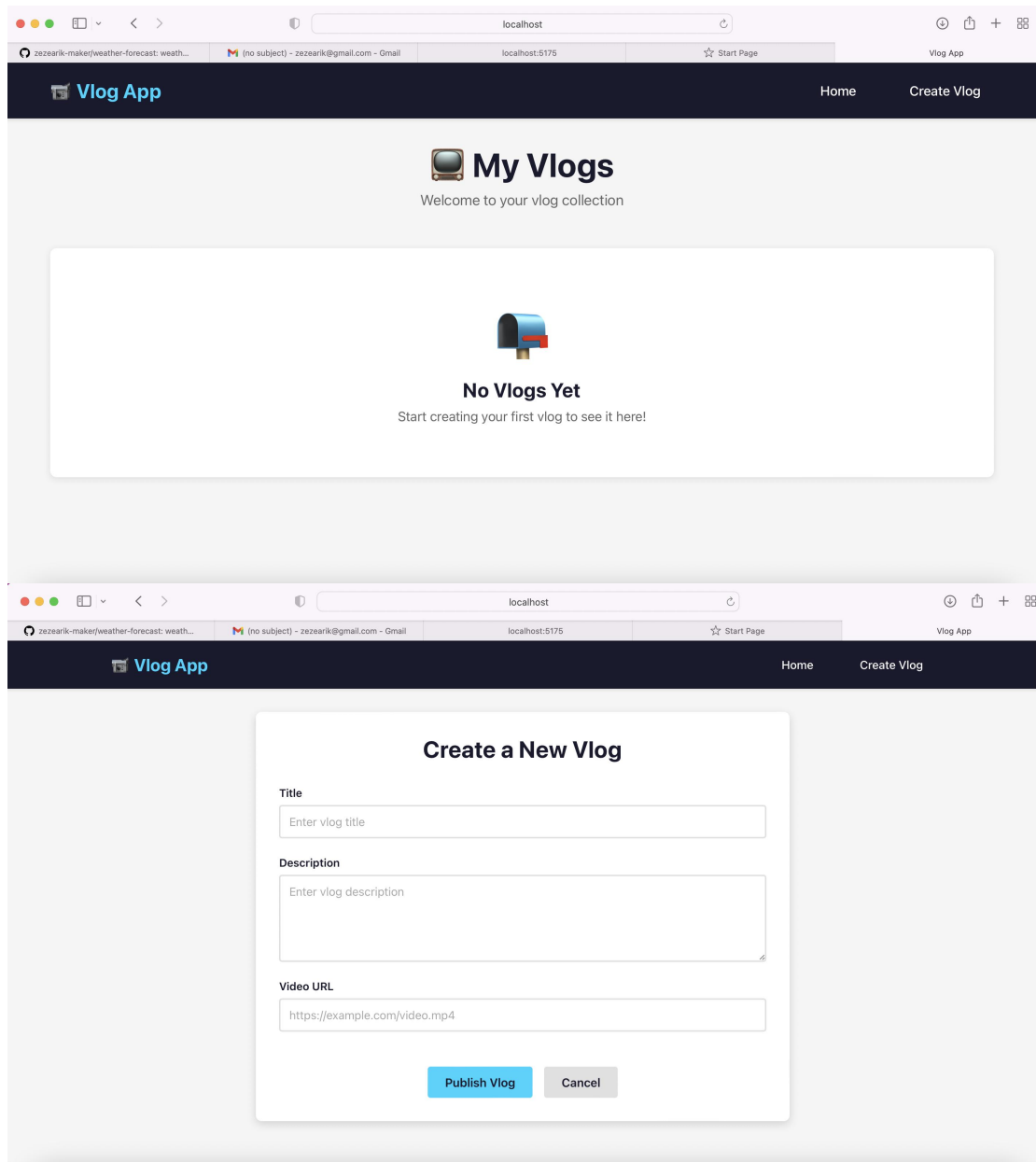
- Frontend Framework: React JS (v18.2.0)
- Routing Library: React Router DOM (v6.20.0)
- Build Tool: Vite (v5.0.8)
- Runtime Environment: Node.js
- Code Editor: Visual Studio Code
- Styling: CSS3 (Responsive Design)
- Languages: HTML, CSS, JavaScript (ES6+)
- Data Storage: LocalStorage API
- Package Manager: NPM

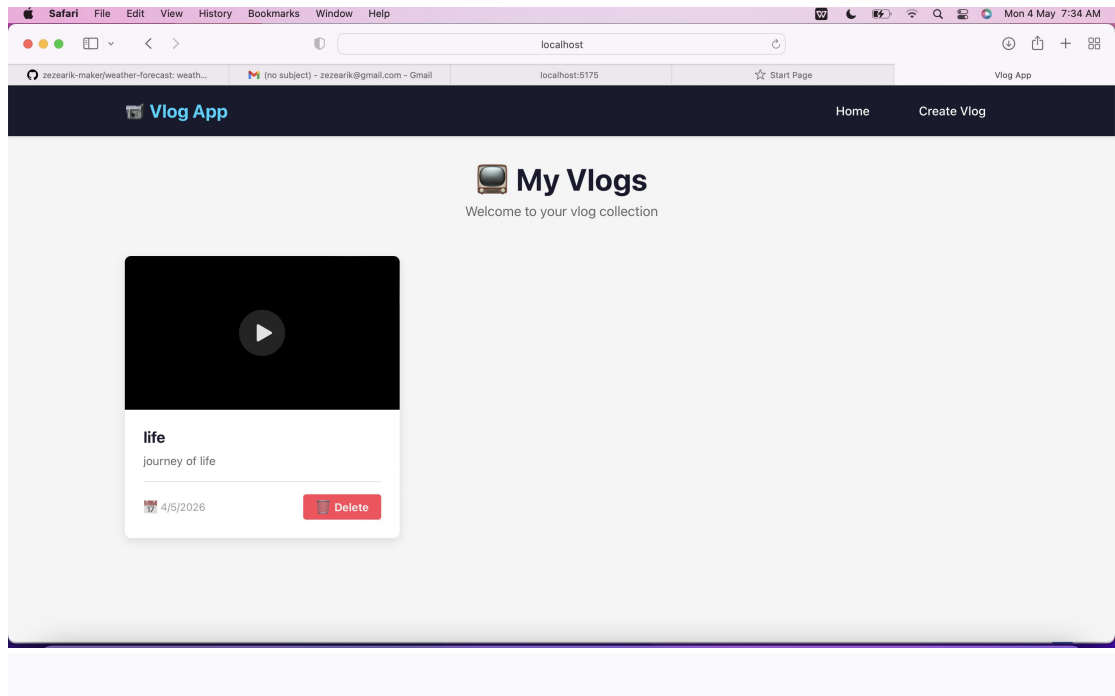
PROJECT DESCRIPTION

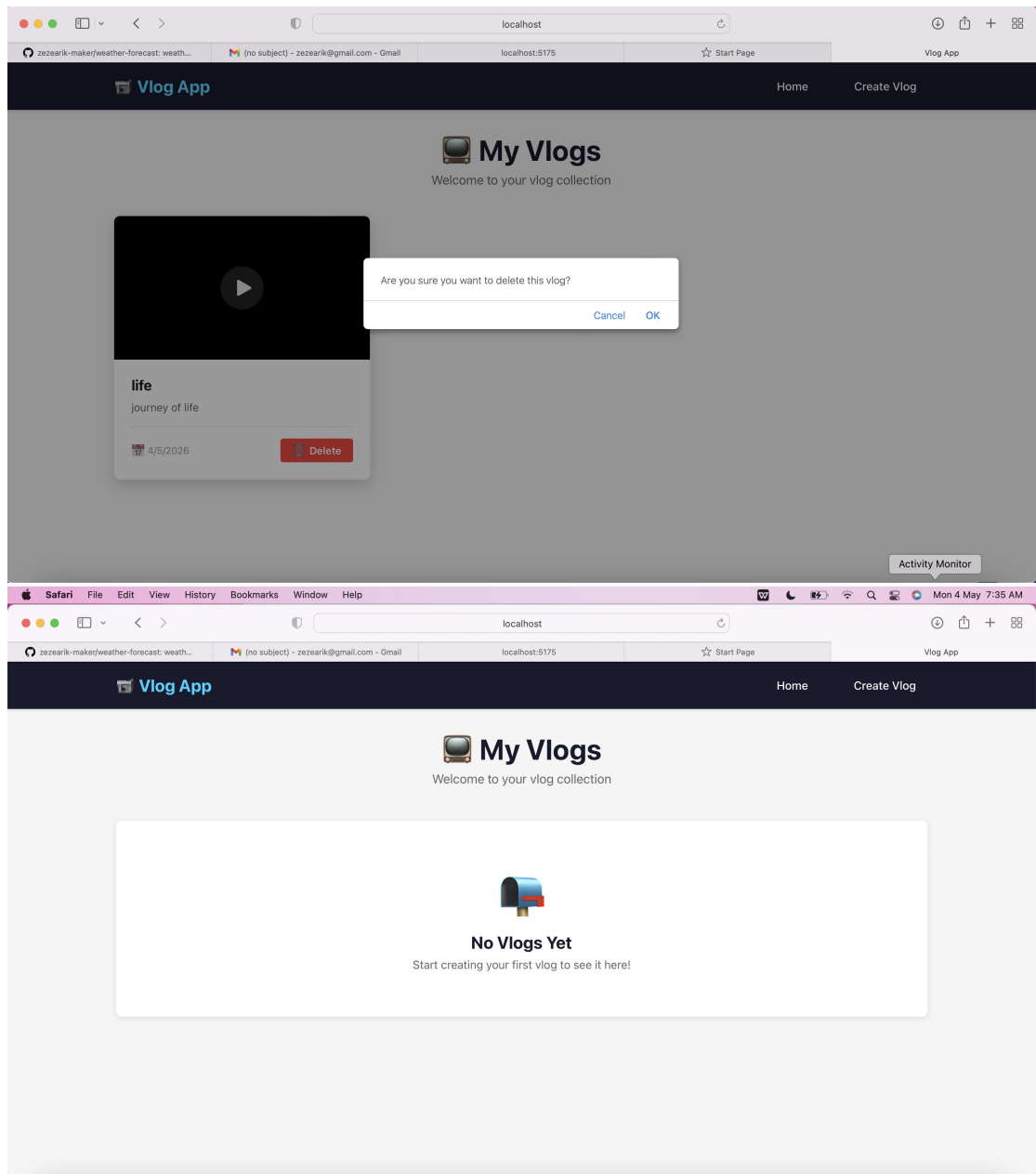
The Vlog Application is a single-page application (SPA) built with React and React Router that allows users to create, view, and delete vlog (video blog) posts. The application provides a user-friendly interface with smooth navigation between pages.

Key Features:

- Home Page - Displays all created vlogs in a responsive grid layout with video players, titles, descriptions, and delete functionality.
- Create Vlog Page - A form that allows users to add new vlogs by entering title, description, and video URL.
- React Router Navigation - Seamless navigation between Home and Create Vlog pages using React Router v6.
- Delete Functionality - Users can delete vlogs with a confirmation dialog to prevent accidental deletion.
- LocalStorage Persistence - All vlogs are automatically saved to the browser's local storage and persist across sessions.
- Responsive Design - The application is fully responsive and works perfectly on desktop and mobile devices.
- Empty State - Shows a friendly empty state message when no vlogs exist.







CODE SECTION

Key Code Snippets:

1. App.jsx - Main Component with State Management

```
import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
```

```
import { useState, useEffect } from 'react';
```

```
import Navigation from './components/Navigation';
```

```
import HomePage from './pages/HomePage';
```

```
import CreateVlogPage from './pages/CreateVlogPage';
```

```
function App() {
```

```
  const [vlogs, setVlogs] = useState([]);
```

```
  // Load vlogs from localStorage on mount
```

```
  useEffect(() => {
```

```
    const savedVlogs = localStorage.getItem('vlogs');
```

```
    if (savedVlogs) {
```

```
      setVlogs(JSON.parse(savedVlogs));
```

```
    }
```

```
  }, []);
```

```
  // Save vlogs to localStorage whenever they change
```

```
  useEffect(() => {
```

```
    localStorage.setItem('vlogs', JSON.stringify(vlogs));
```

```
  }, [vlogs]);
```



```
const addVlog = (vlog) => {

  const newVlog = {

    id: Date.now(),

    ...vlog,

    createdAt: new Date().toLocaleDateString(),

  };

  setVlogs([newVlog, ...vlogs]);

};

const deleteVlog = (id) => {

  setVlogs(vlogs.filter(vlog => vlog.id !== id));

};

return (

  <Router>

    <Navigation />

    <Routes>

      <Route path="/" element={<HomePage vlogs={vlogs} onDelete={deleteVlog} />} />

      <Route path="/create" element={<CreateVlogPage onAdd={addVlog} />} />

    </Routes>

  </Router>

);

}

export default App;
```

Explanation: This is the root component that manages the entire application state. It uses React hooks (useState, useEffect) to manage vlogs and localStorage. The Router component enables client-side routing, and Routes defines two main pages: Home and Create. The addVlog and deleteVlog functions handle state updates.

2. CreateVlogPage.jsx - Form Component

```
import { useState } from 'react';

import { useNavigate } from 'react-router-dom';

export default function CreateVlogPage({ onAdd }) {

  const [title, setTitle] = useState("");

  const [description, setDescription] = useState("");

  const [videoUrl, setVideoUrl] = useState("");

  const navigate = useNavigate();

  const handleSubmit = (e) => {

    e.preventDefault();

    if (!title.trim() || !description.trim() || !videoUrl.trim()) {

      alert('Please fill in all fields');

      return;

    }

    onAdd({

      title,
```

```
description,

    videoUrl,

  });

  setTitle("");

  setDescription("");

  setVideoUrl("");

  navigate('/');

};

return (

  <div className="create-vlog-container">

    <div className="create-vlog-card">

      <h1>Create a New Vlog</h1>

      <form onSubmit={handleSubmit} className="vlog-form">

        <div className="form-group">

          <label htmlFor="title">Title</label>

          <input

            type="text"

            id="title"

            value={title}

            onChange={(e) => setTitle(e.target.value)}

            placeholder="Enter vlog title"

            className="form-input"

          />

        </div>

      </form>

    </div>

  </div>

);
```

```

    </div>

    { /* Similar for description and videoUrl */ }

    <div className="form-actions">

        <button type="submit" className="btn btn-primary">

            Publish Vlog

        </button>

        <button type="button" onClick={() => navigate('/') } className="btn btn-secondary">

            Cancel

        </button>

    </div>

</form>

</div>

</div>

);
}

```

Explanation: This component handles vlog creation. It maintains local state for form inputs using useState hook. The handleSubmit function validates inputs and calls the onAdd prop callback. The useNavigate hook redirects users to the home page after successful submission.

3. HomePage.jsx - Display and Delete Component

```

export default function HomePage({ vlogs, onDelete }) {

    return (

        <div className="home-container">

            <div className="home-header">

```

```

<h1>3 My Vlogs</h1>

<p>Welcome to your vlog collection</p>

</div>

{vlogs.length === 0 ? (

  <div className="empty-state">

    <div className="empty-icon">ô</div>

    <h2>No Vlogs Yet</h2>

    <p>Start creating your first vlog to see it here!</p>

  </div>

) : (

  <div className="vlogs-grid">

    {vlogs.map((vlog) => (

      <div key={vlog.id} className="vlog-card">

        <div className="vlog-video-container">

          <video controls className="vlog-video">

            <source src={vlog.videoUrl} type="video/mp4" />

          </video>

        </div>

        <div className="vlog-content">

          <h3>{vlog.title}</h3>

          <p className="vlog-description">{vlog.description}</p>

          <div className="vlog-footer">

            <span className="vlog-date">ã {vlog.createdAt}</span>

            <button

```

```

        onClick={() => {
            if (confirm('Are you sure you want to delete this vlog?')) {
                onDelete(vlog.id);
            }
        }}

        className="btn-delete"

    >

        Delete

    </button>

</div>

</div>

</div>

    )}

</div>

});
}

```

Explanation: This component displays all vlogs in a responsive grid layout. It uses the `map()` function to iterate over vlogs and render individual cards. The ternary operator shows an empty state when no vlogs exist. The delete functionality includes a confirmation dialog using the `confirm()` method.

4. Navigation.jsx - Route Navigation

```
import { Link } from 'react-router-dom';

export default function Navigation() {

  return (

    <nav className="navbar">

      <div className="nav-container">

        <Link to="/" className="nav-logo">

          📺📺 Vlog App

        </Link>

        <div className="nav-menu">

          <Link to="/" className="nav-link">Home</Link>

          <Link to="/create" className="nav-link">Create Vlog</Link>

        </div>

      </div>

    </nav>

  );

}
```

Explanation: This component provides navigation using React Router's Link component. It renders two navigation links for Home and Create Vlog pages. Links don't cause full page reloads, maintaining the SPA experience.

Project Structure:

```
src/  
├── components/  
│   ├── Navigation.jsx  
│   └── Navigation.css  
├── pages/  
│   ├── HomePage.jsx  
│   ├── HomePage.css  
│   ├── CreateVlogPage.jsx  
│   └── CreateVlogPage.css  
├── App.jsx  
├── App.css  
├── main.jsx  
└── index.css  
vite.config.js  
package.json  
index.html
```


LEARNING OUTCOMES

- React Fundamentals - Understood component-based architecture and how to build reusable UI components using functional components and hooks.
- React Hooks - Mastered `useState` for state management and `useEffect` for side effects like `localStorage` synchronization.
- React Router - Implemented client-side routing with React Router v6, understanding the difference between navigation links and standard anchor tags.
- State Management - Learned to lift state up in React and pass data through props to maintain a single source of truth.
- LocalStorage API - Implemented data persistence using browser `localStorage`, enabling vlogs to survive page refreshes.
- Responsive Design - Created mobile-friendly CSS using CSS Grid, media queries, and flexbox for responsive layouts.
- Form Handling - Implemented form submission with validation and error handling.
- ES6+ JavaScript - Used modern JavaScript features like arrow functions, template literals, destructuring, and spread operators.

CONCLUSION

The Vlog Application successfully demonstrates the core concepts of React development including component architecture, state management, routing, and data persistence. The application provides a seamless user experience with intuitive navigation, form validation, and data safety features. This project has been instrumental in understanding how modern SPAs work and the importance of user-centric design. The responsive layout ensures accessibility across all devices. With the foundation established, the application can be extended with features like user authentication, video uploads, likes/comments, and backend API integration. This assignment has significantly enhanced our understanding of React as a powerful framework for building interactive web applications.

REFERENCES

- React Official Documentation: <https://react.dev>
- React Router Documentation: <https://reactrouter.com/>
- Vite Documentation: <https://vitejs.dev/>
- MDN Web Docs - LocalStorage: <https://developer.mozilla.org/en-US/docs/Web/API/Window/localStorage>
- CSS Grid Layout Guide: https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Grid_Layout
- React Hooks Documentation: <https://react.dev/reference/react>
- ES6 Features: <https://www.ecma-international.org/ecma-262/>
- Web Video Standards: <https://html.spec.whatwg.org/multipage/media.html>